Jeremy **Nuttall**

Principal Software Engineer & Tech Lead

jeremy@jeremy-nuttall.com | jeremy-nuttall.com | jtnuttall | jeremy-nuttall

Experience

Principal Software Engineer & Tech Lead

June 2024–Present

Well.co (healthcare engagement platform) *Remote*

- Took ownership of the services platform for a healthcare engagement platform serving 400K+ members across Bank of America, USAA, Sam's Club, Regions Bank, and UNC Health: Haskell services on ECS Fargate (~480K lines), 21 TypeScript services on Lambda, ~80K lines of Terraform, and GitLab CI pipelines.
- Drove a fiscal-year-critical reward vendor migration (~3 months), guiding architecture and offshore implementation with zero member-visible disruption—delivering $1M+ annualized savings.
- Audited cloud infrastructure across DynamoDB, KMS, and S3, identifying duplicate KMS keys ($100K+/yr), abandoned cross-region redundancy, missing TTL policies, and uncleared retention buckets—achieving $250K+ annualized cost reductions with further remediation ongoing.
- Redirected a planned monolithization toward a monorepo approach, presenting the case to C-level leadership; built the GitLab CI pipeline with Turborepo caching and conditional child pipelines—cutting pipeline times 60%—and authored a migration playbook the team used to unify 21 services in 2 months.
- Hardened authentication and API security handling 2M+ monthly login events; led external security audits and penetration tests with Bank of America, eliminated timing side-channels, implemented MFA and compromised credential checking, and cleared all audits. Cut MTTR to under one hour with proactive Datadog/PagerDuty monitoring.
- Delivered enterprise SSO (SAML 2.0, Amazon Cognito) end-to-end, scoping requirements with C-level stakeholders to close B2B deals with Regions Bank and USAA; 75% of onboarded users adopted SSO, and reusable runbooks halved subsequent onboarding time.
- Grew the team from 6 to 10, personally selecting the engineering manager and 4 of 8 senior engineers; built the leadership bench by advocating for senior engineers to take workstream ownership.

Senior Software Engineer

September 2022–May 2024

Well.co (healthcare engagement platform) *Remote*

- Spearheaded the migration of 200K+ LoC from JavaScript to TypeScript, defining the iterative conversion strategy and selecting Zod for runtime validation—driving a 40% reduction in runtime exceptions and accelerating subsequent initiatives like monorepo adoption.
- Pinpointed backend performance bottlenecks via APM data (p90/p99 latency, bundle sizes), shipping an esbuild integration that reduced Lambda bundle sizes by up to 99.8% across 21 microservices.
- Trained 5 engineers on TypeScript through targeted pairing and progressive code reviews; authored the team's coding standards and linting rules—enabling strict typing adoption, catching type-safety violations at build time, and unblocking a stalled feature backlog.

Senior Software Engineer

November 2021–September 2022

Triplebyte *Remote*

- Owned test integrity end-to-end for Triplebyte's 250K+ engineer assessment platform—building duplicate-account detection with weighted candidate ranking, incognito-mode detection via browser fingerprinting, and a Winnowing-based plagiarism engine (the algorithm behind Stanford's MOSS)—achieving zero false positives across all flagged submissions.
- Built and shipped the employer activation flow for the Screen product: a 4-step onboarding checklist, pre-built assessment templates (Instant Tests), and a guided test-creation wizard covering 9 role types across 3 seniority levels, improving employer activation completion by ~15%.
- Instrumented Screen's test-creation, team-invite, and ATS-integration funnels via Segment, giving the CPO visibility into conversion drop-off across the employer activation pipeline; integrated Salesforce tracking to route enterprise demo requests into CRM.

Software Engineer (Project Lead)

June 2021–November 2021

Appstem *Remote*

- Led project delivery as primary client-facing technical contact, shipping a membership-based React Native application (Google Firebase backend) from scratch to production; simultaneously delivered e-commerce features for KFC's mobile application as a subcontractor.

Software Engineer

July 2019–June 2021

eQ Technologic *Costa Mesa, CA / Remote*

- Reverse-engineered an undocumented enterprise PLM platform to deliver 20+ client-facing web applications for government and defense contractors, operating under strict browser-compatibility constraints with no transpilation toolchain.
- Built HIPAA-compliant backends using Spring (Java) and Oracle PL/SQL for an internal enterprise healthcare data management platform.

Skills

Languages	Haskell, TypeScript/JavaScript, Ruby, SQL
Frontend	React, React Native, Next.js, Redux, GraphQL
Backend	Node.js, Fastify, Ruby on Rails, OAuth 2.0/OIDC, SAML 2.0
Cloud	AWS (Lambda, ECS, Cognito, S3, SQS, SNS, Kinesis), Terraform, Docker
Data	PostgreSQL, DynamoDB, MongoDB, Redis, Elasticsearch
DevOps	GitLab CI, GitHub Actions, Turborepo, Datadog, Linux, Nix

Open Source

pure-noise

Haskell

Author & Maintainer *Hackage*

- High-performance noise generation library achieving 84–95% of C++ FastNoiseLite performance through LLVM-targeted optimization. Features algebraic composition of noise functions via standard Haskell abstractions with minimal dependencies.

heph-aligned-storable

Haskell

Author & Maintainer *Hackage*

- Generic derivation of Storable instances conforming to GPU memory layout specifications (std140/std430/scalar) for safe, high-performance CPU-GPU transfers in a custom rendering engine.

Education

University of Southern California

B.S. in Computer Science *Los Angeles, CA*